

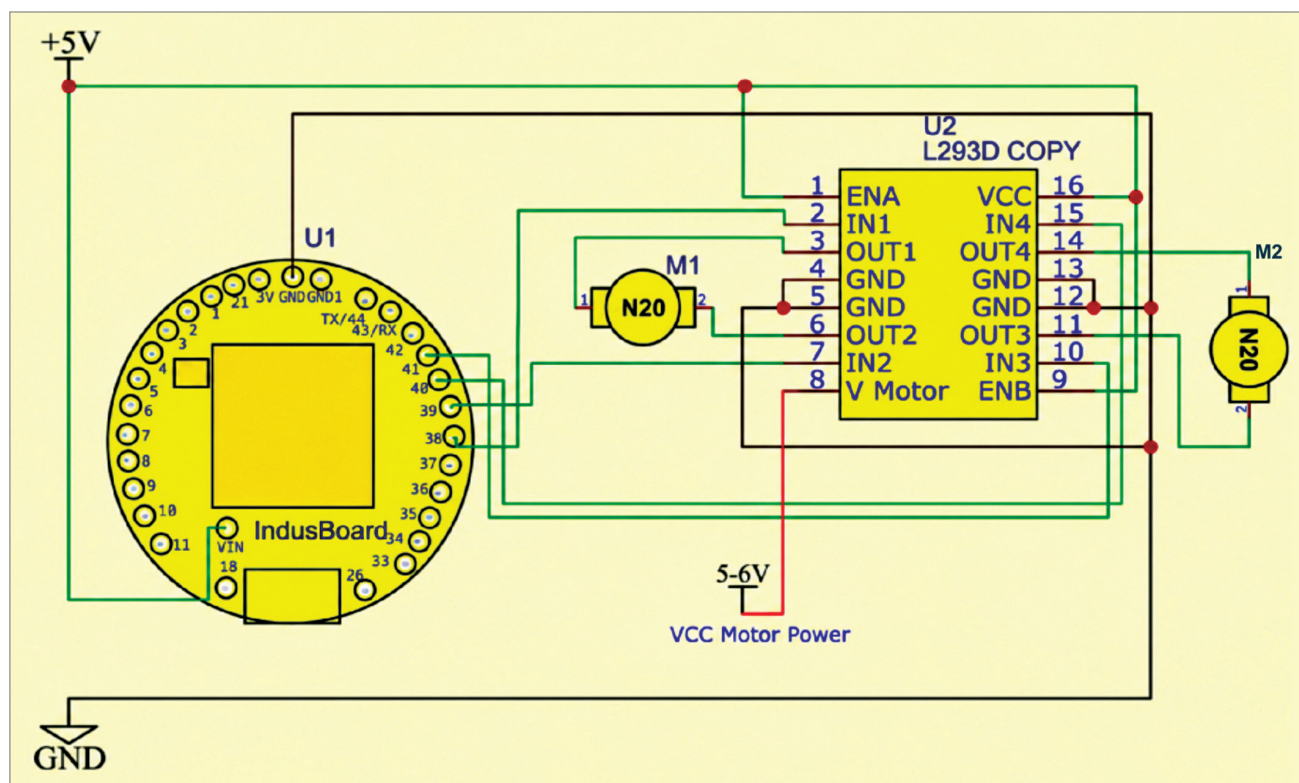


Fig. 6: IndusBoard Coin and L293D motor interface

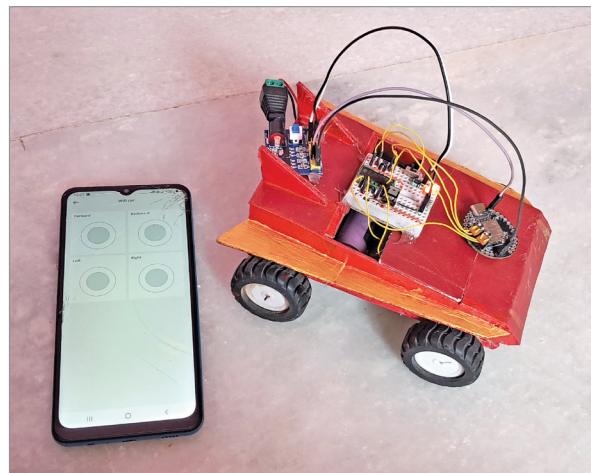


Fig. 7: Testing setup with mobile phone

continues rotating.

The ESP32's built-in Wi-Fi capability allows remote control of the robot through the Arduino IoT Cloud interface from anywhere with internet access.

Construction and testing

First, upload the Arduino IoT Cloud program to the IndusBoard Coin and ensure that it compiles and uploads

successfully.

Connect the L293D logic supply to +5V and the motor supply to a 6V battery, ensuring a common ground connection throughout the circuit.

Securely mount the geared motors, wheels, battery holder, IndusBoard Coin, and L293D module to the robot chassis, and arrange the wiring neatly. The completed system is

shown in Fig. 1.

For testing, allow the IndusBoard Coin to connect to the configured Wi-Fi network. Open the Arduino IoT Cloud dashboard on a smartphone or computer and confirm that the device is online.

Use the Forward, Backward, Left, and Right controls to verify the robot's movement. If a motor rotates in the wrong direction,

interchange its terminals. Perform a final movement test on a flat surface to ensure smooth operation in all directions.

The robot can be controlled from the Arduino IoT Cloud dashboard using a web browser or the Arduino IoT remote mobile app. Table 2 shows the dashboard controls and their corresponding functions.

The dashboard provides four push-button controls for robot navigation. The forward and backward buttons move the robot forwards and backwards, respectively, while the left and right buttons control turning. The corresponding IndusBoard Coin pin combinations are listed in Table 2. These controls enable wireless operation of the robot through the Arduino IoT Cloud platform. Fig. 7 shows the testing setup with the prototype and mobile phone. **EFY**



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